ELEC40001 - MATHEMATICS: PRINCIPLES AND ENGINEERING APPLICATIONS

1. a) Express in the form $x+iy$:

$$(i)\quad \left(\frac{3i-2}{2i+3}\right)^2, \qquad (ii)\quad \left(\frac{i-1}{2}\right)^{2023}, \quad (iii)\quad (-1)^i, \qquad [\,6\,]$$

b) (i) Obtain the real and imaginary parts of $\cos(iz)$; [3]

(ii) Hence, or otherwise, obtain all complex z satisfying the equation

$$\cos(iz) = -i\,. \qquad [\,4\,]$$

c) Sketch the function $f(t) = t(2+t)[H(t+1)-H(t)]$, where $H(t)$ is the Heaviside function. Hence sketch $f_e(t)$ and $f_o(t)$, the even and odd parts of $f(t)$, taking care to explain your reasoning. [4]

d) Obtain the limits [4]

$$(i)\ \lim_{x\to\pi/2}\frac{\cos x}{1+\cos(2x)}. \quad (ii)\ \lim_{x\to\pi/2}\frac{\cot^2 x}{1-\sin x}.$$

Explain your reasoning!

e) Using the definition of derivative as limit, or differentiation from first principles, show that [4]

$$\frac{d}{dx}(\cos x) = -\sin x\ .$$

[You may quote the result for $\lim_{x\to 0}\dfrac{\sin x}{x}$.]

2. a) Differentiate to obtain $\frac{dy}{dx}$: [6]

$$(i) \quad y = (\sin x)^x, \qquad (ii) \quad x^3\sqrt{1-y} - e^{x-y} = 42.$$

b) Given $y = \left(\sin^{-1} x\right)^2$, show that $\sqrt{1-x^2}\frac{dy}{dx} = 2\sin^{-1} x$.

Hence, or otherwise, show that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 2 = 0$. [4]

c) Evaluate the integrals: [5]

$$(i) \qquad \int \frac{\sec^2 x}{\tan x}\, dx\,, \quad (ii) \qquad \int_0^1 x\tan^{-1} x\, dx\,.$$

d) Given that $x^3 + x^2 + x + 1 = 0$ has a zero at $x = -1$, use partial fractions to obtain the indefinite integral [5]

$$\int \frac{2}{1+x+x^2+x^3}\, dx.$$

e) Use a substitution to integrate $\frac{1}{\sqrt{x^2+1}}$ and hence, or otherwise, show that [5]

$$\sinh^{-1} x = \ln(x + \sqrt{x^2+1}).$$

3. a) Use the binomial theorem to find the power series for $y = \dfrac{1}{\sqrt{1-x^2}}$.

Hence, or otherwise, find the first four non-zero terms of the series for $\sin^{-1} x$. [5]

b) Obtain the Taylor series of

$$y = \frac{\ln(x-1)}{x}$$

about $x = 2$, up to the linear term, and give the remainder term R_1. [5]

c) The following solution of a first-order ordinary differential equation (ODE) is incorrect.

The ODE $x\dfrac{dy}{dx} = y\ln\left(\dfrac{y}{x}\right) + y$ *is homogeneous.*

Hence, it can be written as $\left(\dfrac{1}{z\ln z}\right)\dfrac{dz}{dx} = \dfrac{1}{x}$ *where* $z = y/x$.

This is in separable form which gives $\ln(\ln(z)) = \ln(x)$,

and hence the solution $y = Axe^x$, *where A is an arbitrary constant of integration.*

Which was the **first** error made in this calculation? Which was the second error? Now solve the ODE correctly. [6]

d) The second-order ordinary differential equation

$$(1-x^2)\frac{d^2y}{dx^2} + 2y = 0$$

can be solved with a series solution. Differentiate the ODE n times, using the Leibniz formula. Show that

$$y^{(n+2)}(0) = (n-2)(n+1)y^{(n)}(0)$$

for $n \geq 0$, where $y^{(n)}(0)$ is the nth derivative of y, evaluated at $x = 0$. Given initial conditions $y(0) = 0$ and $y'(0) = -1$, show that

$$y = -x + \frac{1}{3}x^3 + \frac{1}{15}x^5 + \frac{1}{35}x^7 + \frac{1}{63}x^9 + \ldots$$ [9]

4. a) Solve the second-order ordinary differential equation

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} - 3x = 4e^{-t}$$

satisfying the initial conditions $x = 0$, and $\dfrac{dx}{dt} = 1$, when $t = 0$. [7]

b) A regular cylinder of radius r and height h has volume $V = \pi r^2 h$. Suppose the volume is to be approximated by estimating radius and height. What is the maximum (equal) error we can incur in estimating radius and height, if the error in the volume is to be at most 0.3%? [5]

c) Polar coordinates (r, θ) in are related to cartesian coordinates (x, y) as follows:

$$x = r\cos\theta \ , \qquad y = r\sin\theta \ ,$$

For a function $f(x,y)$, obtain $\dfrac{\partial f}{\partial r}$ and $\dfrac{\partial f}{\partial \theta}$ in terms of $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

Hence, or otherwise, show that

$$\frac{\partial f}{\partial x} \ = \ \cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta} \quad \text{and} \quad \frac{\partial f}{\partial y} \ = \ \sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r}\frac{\partial f}{\partial \theta},$$

for $r > 0$. [8]

d) Sketch the region of integration and change the order of integration to solve the double integral [5]

$$\int_0^6 \int_{x/3}^2 x\sqrt{y^3+1}\, dy\, dx \, .$$